Working summary: Holocene climate, fire and human-use intervals for NOCA and MORA

Purpose: define defensible event windows for the climate-community lag analysis, while keeping climate, fire and archaeology as independent evidence streams.

Decision rules used here

- Dates are expressed as calibrated years BP (cal yr BP) whenever the source supports calibrated chronology.
- Older conventional radiocarbon ages are not placed directly on a cal-BP axis unless independently calibrated.
- Climate windows should be selected from climate evidence, fire windows from charcoal/fire evidence, and human-use windows from archaeology. Vegetation changes should remain responses, not criteria used to choose the window.
- MCA and LIA are retained as regional context, but are not currently prioritized as core local climate events because the CHELSA-TraCE21k trajectories do not show uniquely strong local excursions.

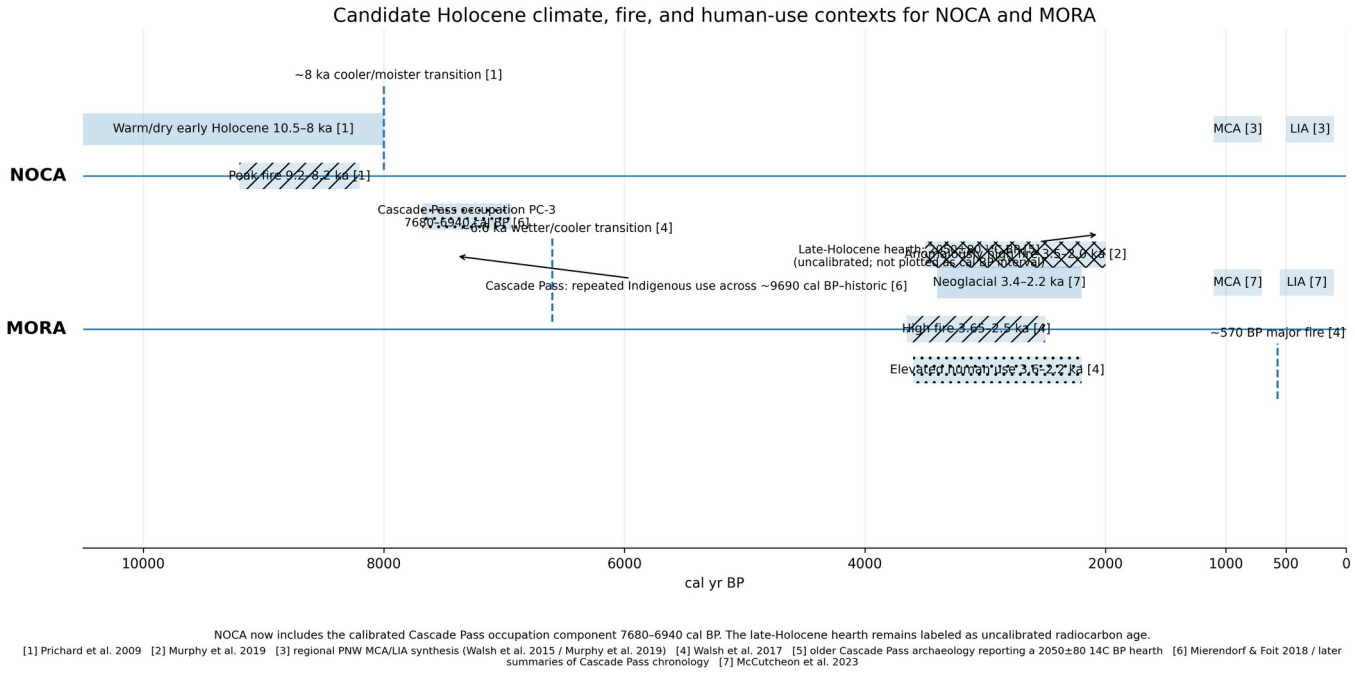


Figure 1. Working timeline of candidate climate, fire and human-use contexts. The NOCA 7680-6940 cal BP occupation is a calibrated archaeological component; the 2050 +/- 80 14C BP hearth remains explicitly uncalibrated.

1. NOCA - North Cascades / Panther Potholes

Interval	Evidence type	Interpretation / use	CHELSA-TraCE match	Ref.
10,500-8,000 cal BP	Climate	Warm/dry early-Holocene regime. Strong local Panther Potholes period; keep as a core NOCA climate context.	Good broad match; early Holocene is climatically dynamic in local TraCE series.	[1]
9,200-8,200 cal BP	Fire	Peak early-Holocene fire frequency at Panther Potholes. Use as a fire overlay within the early-Holocene climate regime.	Not expected to be a separate climate interval; fire peak occurs within dynamic early-Holocene climate.	[1]
~8,000 cal BP	Climate transition	Shift toward cooler/moister conditions. Prefer treating this as a transition zone rather than an instantaneous breakpoint.	Good match to changes in direction across local temperature/precipitation trajectories.	[1]
7,680-6,940 cal BP	Archaeology	Cascade Pass occupation component PC-3. Evidence	Independent of climate selection; useful human-use	[6]

		of calibrated upland Indigenous use; not evidence of region-wide continuous settlement.	context after the ~8 ka transition.	
~9,600 cal BP-historic	Archaeology	Repeated Indigenous use of Cascade Pass across much of the Holocene. Use qualitatively as persistent upland use, not as a continuous high-use interval.	Context only.	[6]
3,500-2,000 cal BP	Fire	Anomalously high late-Holocene fire activity at Panther Potholes relative to many regional records. Strong disturbance-test window.	Climate changes, but this interval is not uniquely extreme in local CHELSA-TraCE; therefore useful as a disturbance exception.	[2]
2,050 +/- 80 14C BP	Archaeology	Cultural hearth reported from Cascade Pass. Do not plot as cal BP until calibrated from the original determination.	Not used as a calibrated interval.	[5]
1,100-700 cal BP	Regional climate	Medieval Climate Anomaly (MCA). Retain as regional PNW context only.	Weak/moderate local expression; not uniquely prominent in local CHELSA-TraCE.	[3]
~500-100 cal BP	Regional climate	Little Ice Age (LIA). Retain as regional PNW context only.	Weak local expression; 500-yr smoothing can blur it, but even absolute/finer trajectories do not show a uniquely strong local event.	[3]

2. MORA - Mount Rainier

Interval	Evidence type	Interpretation / use	CHELSA-TraCE match	Ref.
~6,600 cal BP	Climate transition	Transition toward wetter and eventually cooler conditions; modern mesic forest/subalpine meadow landscapes become established around this time in Walsh synthesis. Core MORA climate transition.	Very good match: MORA pond trajectories show coordinated climate reorganization around ~7-6.5 ka.	[4]
3,650-2,500 cal BP	Fire	Strong high-fire interval at Sunrise Ridge, between MSH-Yn and MSH-Pu tephras. Strong disturbance-test window.	Climate is changing but not uniquely extreme; supports testing fire as a modifier rather than simply climate magnitude.	[4]
3,600-2,200 cal BP	Archaeology	Elevated archaeological activity / human use at Sunrise Ridge, based on high artifact density and diversity. Strongest local human-use overlap with fire.	Independent disturbance context.	[4]
3,400-2,200 cal BP	Climate	Neoglacial cool/wet interval in the updated MORA synthesis. Useful background climate context overlapping the fire/human-use episode.	Moderate-good: expressed more as continued cooling/variability than a single abrupt event.	[7]
1,100-700 cal BP	Regional climate	MCA. Retain as regional context rather than a primary local event.	Weak/moderate local expression in CHELSA-TraCE.	[7]
550-100 cal BP	Regional climate	LIA. Retain as regional context rather than a primary local event.	Weak local expression in CHELSA-TraCE.	[7]
~570 cal BP	Fire	Large local fire episode in the Shadow Lake record. Potential individual-event annotation within the late Holocene/LIA-age context.	Useful event test, not a climate interval.	[4]

3. Recommended analytical priority

- NOCA core climate contrasts: 10.5-8 ka early-Holocene warm/dry regime and the ~8 ka transition toward cooler/moister conditions.
- NOCA disturbance test: 3.5-2.0 ka anomalously high fire. Treat long-term Indigenous use as context; do not currently assign a 3.5-2.0 ka "high human-use" treatment.
- MORA core climate contrast: ~6.6 ka wetter/cooler transition; use 3.4-2.2 ka Neoglacial conditions as late-Holocene climate background.
- MORA disturbance test: 3.65-2.5 ka high fire, explicitly considered alongside 3.6-2.2 ka elevated archaeological activity.
- MCA/LIA: retain as regional annotations and sensitivity analyses, not primary event windows unless a finer-resolution local climate analysis identifies a distinct event signal.

CHELSA-TraCE21k cross-check

The local climate files show that the strongest literature-defined periods are broadly compatible with the site-specific climate reconstruction, but not every named regional episode emerges as a discrete local event. The absolute Holocene trajectories show a strong long-term summer-temperature cooling signal, whereas MCA/LIA are comparatively subtle. The 500-year change plot can additionally attenuate events lasting only a few centuries. Therefore, event selection should rely on the raw/finer climate trajectory when testing MCA/LIA, while the broad early/mid-Holocene transitions remain well represented. [8-9]

4. Numbered references

[1] Prichard, S.J., Gedalof, Z., Oswald, W.W. & Peterson, D.L. (2009). Holocene fire and vegetation dynamics in a montane forest, North Cascade Range, Washington, USA. Quaternary Research 72: 57-67.

[2] Murphy, S.F. et al. (2019). A 5000-Year Fire History in the Strait of Georgia Lowlands, British Columbia, Canada. Frontiers in Ecology and Evolution 7:90. Used here as the later regional comparison highlighting Panther Potholes as unusual during parts of the late Holocene.

[3] Walsh, M.K., Marlon, J.R., Goring, S.J., Brown, K.J., Gavin, D.G. et al. (2015). A Regional Perspective on Holocene Fire-Climate-Human Interactions in the Pacific Northwest of North America. Annals of the Association of American Geographers 105: 1135-1157.

[4] Walsh, M.K., Lukens, M.L., McCutcheon, P.T. & Burtchard, G.C. (2017). Fire-climate-human interactions during the postglacial period at Sunrise Ridge, Mount Rainier National Park, Washington (USA). Quaternary Science Reviews 177: 246-264.

[5] Older Cascade Pass archaeological reporting of a cultural hearth dated 2050 +/- 80 14C BP. This value is retained explicitly as an uncalibrated radiocarbon determination pending calibration from the original laboratory record.

[6] Mierendorf, R.R. & Foit, F.F. Jr. (2018). Holocene Geochronology and Archaeology at Cascade Pass, Northern Cascade Range, Washington. Journal of Northwest Anthropology, Memoir 16. Later summaries report Indigenous use beginning by ~9,600 cal BP and the calibrated PC-3 occupation at 7680-6940 cal BP.

[7] McCutcheon, P.T., Ferry, J.D., Burtchard, G.C. & Walsh, M.K. (2023). An Approach for Explaining Technological and Functional Variation Between Four Montane Lithic Assemblages Near Mount Rainier, Washington. Journal of Northwest Anthropology 57(2): 167-193.

[8] User climate file: climate_holocene_timeseries.pdf. CHELSA-TraCE21k Holocene change-per-interval trajectories and climate-variable correlation diagnostics.

[9] User climate file: climate_trend_holocene.pdf. Absolute Holocene temperature and precipitation trajectories for MORA and NOCA.

Working-status caution. The event windows above are intended as defensible candidate contrasts, not immutable stratigraphic boundaries. Exact model windows should be aligned to the temporal resolution of the macrofossil data and checked against the unsmoothed/finer CHELSA-TraCE climate trajectories before final analysis.